

The C-PROBE tranSMART instance can be accessed at <https://cprobe-tranSMART.med.umich.edu/>

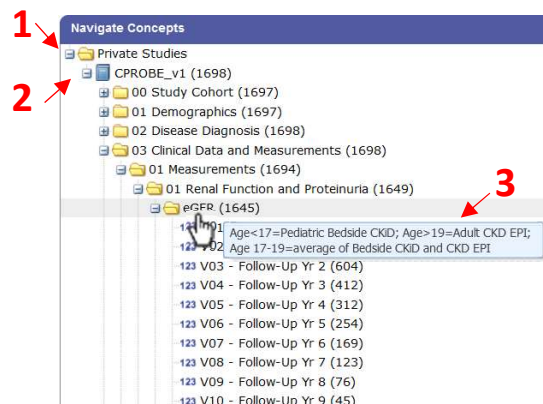
The application works best in Chrome, Firefox, or Safari. It cannot be used from a mobile device. The first time you log in (and once every 90 days after that), you will be asked to attest to a Data Use Statement.

Overview

To explore the data available, click on the plus sign (+) next to “Private Studies” in the **Navigate Concepts** section (1). Then click on the plus sign next to “C-PROBE_v1” to expand the terms tree (2).

The tree provides additional useful information:

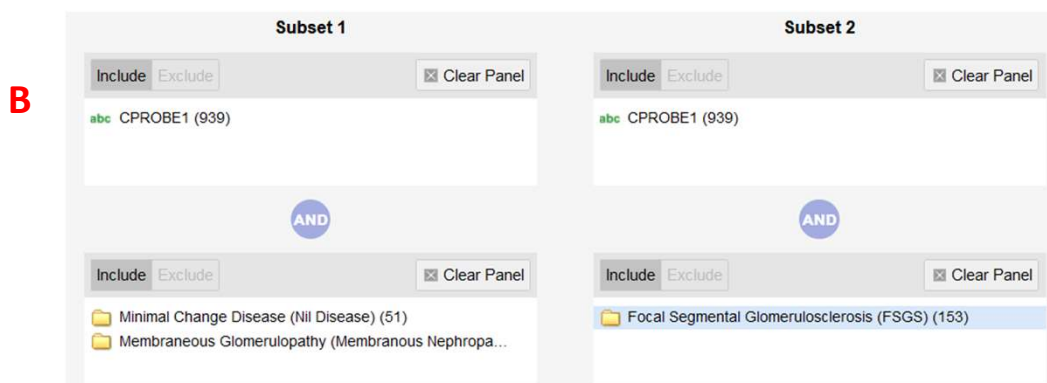
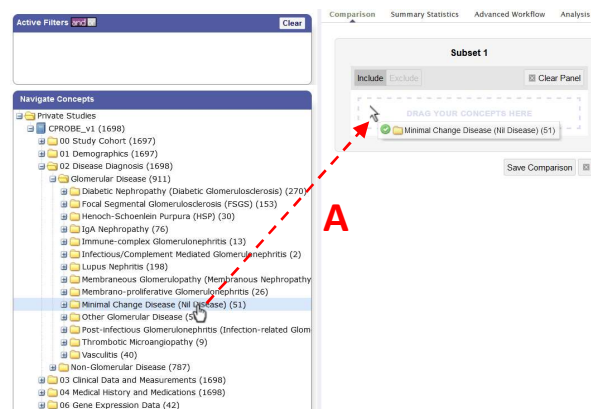
- The numbers in parenthesis after the terms are the number of samples that have that data.
- The **123** or **abc** before a term indicates if the data is numeric or categorical in nature.
- Hovering over a node, or right-clicking on it, will bring up a definition pop-up for some terms (3).



Selecting Data

The first step is to set the **Subset(s)** of data you wish to explore. Drag and drop terms from the tree into the **Subset** boxes on the right (A).

- If an entire folder is dragged over, all samples with a value for that data point are included in the subset regardless of the value (e.g. dragging the folder “Glomerular Disease” would include all patients with any kind of Glomerular Disease).
- If only a single term is included, then only samples having that value are included (e.g. dragging “C-PROBE1” would include only those patients in C-PROBE 1).
- If a numeric term is selected, values can be further restricted by choosing an option from the pop-up menu. Select “No Value” to use all samples with any value.
- Terms in the same subset box are part of an “OR” statement.
- Subset boxes are joined together in “AND” statements.
- Two different subsets (**Subset 1** and **Subset 2**) can be defined and compared. Example B compares **Subset 1**: “C-PROBE1” AND “Minimal Change Disease (Nil Disease)” OR “Membranous Glomerulopathy (Membranous Nephropathy)” to **Subset 2**: “C-PROBE1” AND “Focal Segmental Glomerulosclerosis (FSGS)”.

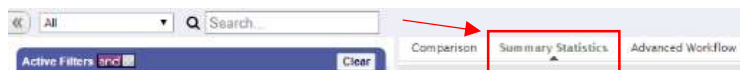


Analyzing Data

After setting a subset, clicking on the **Summary Statistics** tab will bring up an overview of the subset(s) by Age, Sex, and Race. Scroll through the results to see summaries and breakdowns of the terms used in the subset statements. For the subsets created, compare a concept across those Subsets by dragging an additional term from the tree directly onto the **Summary Statistics** results page.

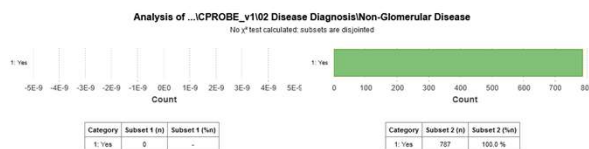
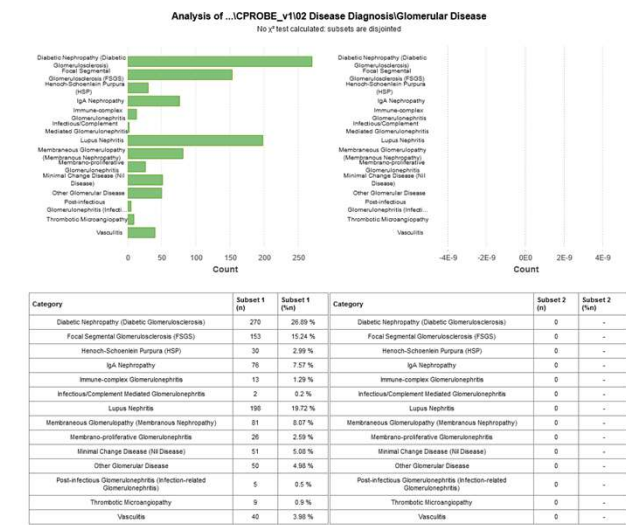
Example 1: Compare eGFR between Glomerular Disease and Non-Glomerular Disease in C-PROBE data.

- From the C-PROBE dataset tree, drag and drop “Glomerular Disease” into **Subset 1** and “Non-Glomerular Disease” into **Subset 2**. Then click on **Summary Statistics**.

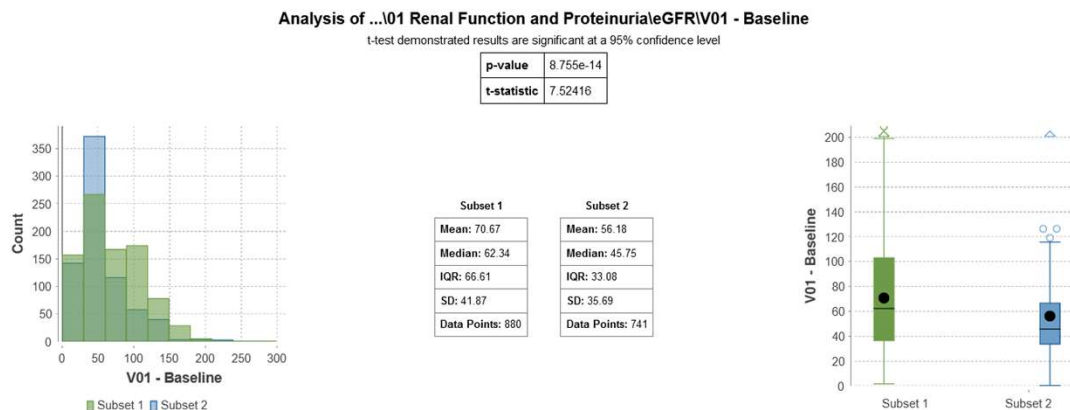


- The top of the Summary Statistics screen shows a comparison of the two subsets:

- The bottom of the Summary Statistics screen shows a breakdown of the counts and values in each subgroup:



- Drag and drop “V01 – Baseline” from the eGFR folder onto the Summary Statistics results page. A box plot comparing eGFR values in each subset will appear at the bottom of the page:



Using Gene Expression Data

Explore genes of interest in gene expression data using the **Advanced Workflow** tab. Use a single gene or upload a gene list to compare multiple genes. To create a gene list, use the **Gene Signature/Lists** link at the top right-hand side of the application (A) and follow the instructions (only fields with an asterisk are required). To use a gene list, type the name of a list into the “Select a Gene” box in the “High Dimensional Data” popup window and select it from the dropdown list (B).

Example 2: Create a scatter plot of eGFR data against Gene Expression levels of COL4A1.

- Subset to desired patients by dragging “Tub” from “Brain array alt CDF v10” folder and “V1 – Baseline” from eGFR folder into the first two Subset boxes on the Comparison tab.
- Click on the Advanced Workflow tab and select “Scatter Plot with Linear Regression” from the “Analysis” menu. (1)
- Drag “Tub” into the “Independent Variable” box. (2)
- Click on the “High Dimensional Data” button under the “Independent Variable” box. (3)
- Type “COL4A1” in the gene box and then select it from the dropdown list. (4) Apply selections.
- Drag “V1 – Baseline” from eGFR folder into the “Dependent Variable” box. (5)
- Click “Run”. (6)

1. Analysis menu: Scatter Plot with Linear Regression

2. Independent Variable: Tub (42)

3. High Dimensional Data button

4. Select a Gene/Pathway/mirID/UniProtID: COL4A1

5. Dependent Variable: V1 - Baseline (1621)

6. Run button

Compare Subsets-Pathway Selection

Marker Type: Gene Expression

GPL Platform: Brain_array_alt_CDF_v10

Sample: Tub

Tissue:

Select a Gene/Pathway/mirID/UniProtID: mygl

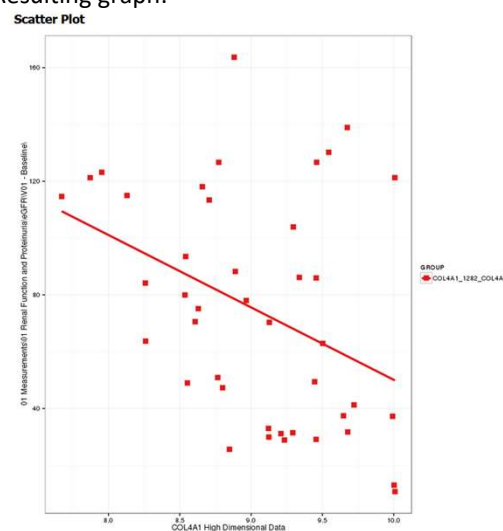
Gene: Myg1 (A1325965, 0610023A07Rik, 281043321Rik, 5830429P19Rik, Gam...)

Gene List: MyGeneList

Gene: Col4a1 (MYG, MST024, MSTP024, Gamm1, MYG1)

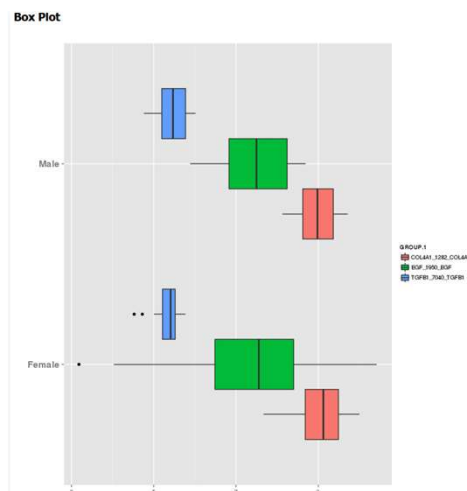
Apply Selections Cancel

Resulting graph:



Example 3: Create a box plot to show expression of genes in Female vs Male.

1. Create a gene list with Col4A1, EGF, and TGFB1 using the **Gene Signatures/Lists** tab as explained above. Then subset to desired patients by dragging the “Sex” folder into the **Subset** box on the **Comparison** tab.
2. Click on the Advanced Workflow tab and select “Box Plot with ANOVA”.
3. Drag “Tub” into the Independent Variable box.
4. Click on the “High Dimensional Data” button.
5. Type the name of your gene list into the gene box and select it from the dropdown menu.
6. Drag “Sex” into the Dependent Variable box.
7. Click “Run”. Resulting graph is shown at right.



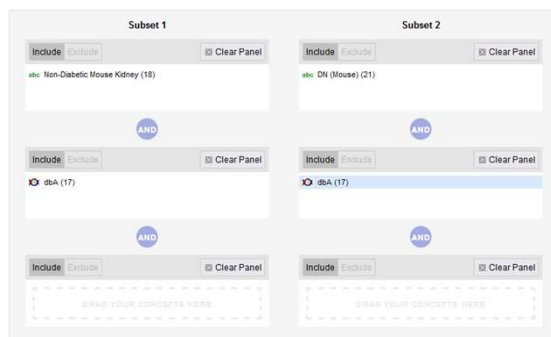
Working with Mouse data

The C-PROBE tranSMART instance includes a node for “ERCB and Mouse data”, which contains gene expression data for 6 mouse models. The metadata for mouse models can be found in the **Navigate Concepts** pane under “04 Mouse Metadata and Measurements”, and the gene expression data is available by model under “06 Gene Expression Data”. (1)

Example 3: Find differential expression of genes between Non-Diabetic Mouse Kidney Control samples and DN (Mouse) samples in the DBA/2 mouse model.

1. From the “ERCB_and_Mouse_data/02 Disease” node of the **Navigate Concepts** tree, drag “Non-Diabetic Mouse Kidney” into Subset 1 and “DN (Mouse)” into Subset 2. Then from under “ERCB_and_Mouse_data/06 Gene Expression Data/Mouse Brain array alt CDF v10”, drag “dbA” into both Subset 1 and Subset 2 (2).

2



2. Click on the Advanced Workflow tab and select “Marker Selection (Differential Expression)”.
3. Drag “dbA” into the variable selection box.
4. Click on the “High Dimensional Data” button. Type 2 or more mouse genes into the “Select a Gene...” box, selecting them from the auto-complete options and separating the genes with a comma. (3)
5. Click Run to view the resulting heatmap. (4)

4

Marker Selection - Heatmap

Click on the heatmap image to open it in a new window as this may increase readability.

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